Notes on GAN

GAN and its architecture:

A Generative Adversarial Network (GAN) is a deep learning model used to generate new data samples that resemble real data.

It consists of two neural networks trained together in a competitive process:

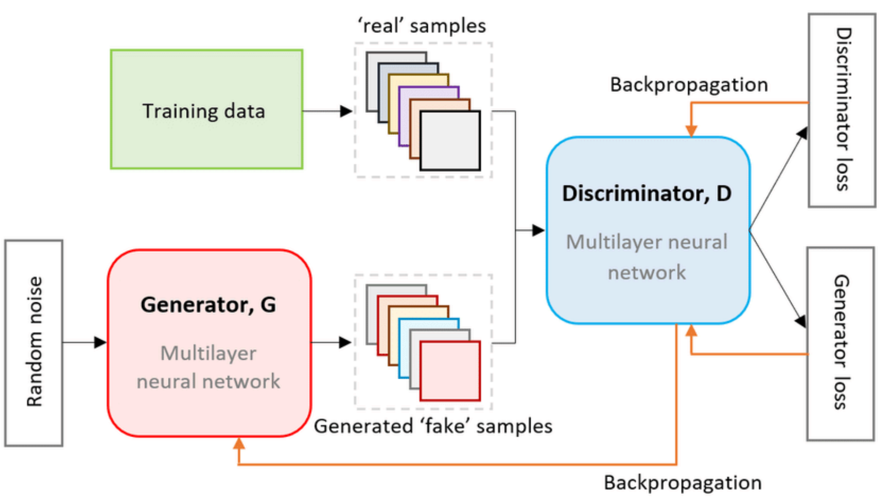
Components of GAN:

- Generator (G)
  - Generates fake data from random noise.
  - Tries to fool the discriminator.
- Discriminator (D)
  - Classifies whether input data is real or fake.
  - Tries to detect generated samples.

Training works like:

- Generator improves to create realistic data.
- Discriminator improves to detect fake data.

This process continues until generated data becomes very realistic.



The different types of GANs and their architecture:

1. DCGAN (Deep Convolutional GAN)

A GAN architecture that uses Convolutional Neural Networks (CNNs) instead of fully connected layers.

Architecture

Generator

- Input: Random noise vector
- Fully connected layer
- Fractional-strided (Transposed) Convolutions
- Batch Normalization
- ReLU activation
- Output image

Discriminator

- Convolution layers
- Batch Normalization
- Leaky ReLU
- Sigmoid output (Real/Fake)

Applications

- Image generation
- Face synthesis
- Feature learning

2. Conditional GAN (cGAN)

GAN where generation is controlled using additional information (condition) such as labels or text.

Architecture

Both Generator and Discriminator receive:

- Random noise z
- Condition y (label/class)

Generator:
Generates images based on condition.

Discriminator:
Checks authenticity + correctness of condition.

Example

- Generate digit “5” from handwritten digits dataset.
- Text-to-image generation.

3. CycleGAN

Used for image-to-image translation without paired datasets.

Example:

- Horse ↔ Zebra
- Summer ↔ Winter images

Architecture

Contains:

- Two Generators
    - G: Domain X → Y
    - F: Domain Y → X
- Two Discriminators
    - Dx and Dy

Key Concept — Cycle Consistency Loss

Ensures:

Original Image ≈ Reconstructed Image
X → Y → X

Applications

- Style transfer
- Domain adaptation
- Image translation

4. SRGAN (Super-Resolution GAN)

GAN is used for image super-resolution, converting low-resolution images into high-resolution images.

Architecture

Generator

- Residual Blocks
- Skip Connections
- Upsampling layers
- Produces high-resolution image

Discriminator

- CNN classifier distinguishing real HR images from generated ones.

Special Feature

Uses **Perceptual Loss** instead of only pixel loss to preserve texture details.

Applications

- Image enhancement
- Medical imaging
- Video upscaling